

combination of file sizes ranging from 1 KB to 100 MB arranged in random order in multiple nested folders) and written onto the CD-R. Speed with absolute quality was the ultimate goal.

Notice how the scores change radically once the writers are submitted to this torture! The rather finicky Asus CRW-2410S suddenly got displaced from its top spot, to be supplanted by the Yamaha CRW 3200E-VK (24/10/40). This writer wrote the entire CD in 226 seconds (3 minutes 46 seconds)! Close on its heels came the Plextor PX-W2410TA, completing the test in a neat 232 seconds (3 minutes 52 seconds). Not far behind was the Iomega CDRW9600INT-B, clocking 238 seconds (3 minutes 58 seconds).

The LG GCE-8160B proved to be the fastest amongst the 16x writers, taking 5 minutes 9 seconds to complete the test, with the Asus CRW 1610A finishing the test in 5 minutes 13 seconds.

Nero's very own file verification test was used to check the data integrity for all the tests. All writers proved to be quite successful in this department and we did not face any data corruption issues while

Is my CD-Writer too Fast for my PC?

CD-Writer speeds are touching over 24x but is your system capable of sustaining those data transfer rates? Does buying a fast CD-Writer make sense when you have a slow performing system? Well, don't be fooled into upgrading just because you bought a fast CD Writer.

For your 'fast' CD-Writer to give you optimal performance, it is sufficient to have a PC with a 500 MHz processor with 128 MB RAM (256 MB is recommended). Though the data transfer rate to the writer is already touching speeds of around 4 MBps it still is much lower than the data throughput a hard drive can achieve. Hence, any 5,400-rpm hard drive should suffice.

BURN GUIDE	
Write Speed	Time required to burn a 650 MB CD-R (optimal)
1x	76 minutes
4x20	minutes
8x	10 minutes
16x	5 minutes
24x	3 minutes

writing at high speed and with an assortment of files.

Cut your own audio CD!

So you want to cut your own audio CD and give it to your friend for his birthday? Not enough time to burn 100 odd CDs at 4x? Well it's not at all a difficult task.

We burnt an audio CD with 14 tracks totalling 64 minutes 2 seconds in DAO (Disk At Once) mode in less than 4 minutes. Also the stakes become a little higher with this test as we checked for jitter, skips and other unwanted noise that may creep into the tracks.

The Yamaha CRW 3200-VK showed its absolute burning prowess by writing

CD Writer Technology: How does it Work?

The fundamentals of CD Writing technology are pretty complex. Just consider the number of factors that come into play when recording a disc: the rate of spin, the formu-

la of the dye, the ambient temperature, the internal temperature, the age of the media, the power and wavelength of the laser, and the spacing and size of the marks on the

media relative to the speed of the disc, to name but a very few. Nevertheless, despite all the variables and compatibility issues, CD-Writing has become extremely reliable.

A CD-Writer in essence is quite simple in construction. It has a motor mounted internally which rotates the media. The laser assembly is mounted on the rack which moves outward from the centre of the media as the media spins.

This assembly is moved precisely to the desired location by a stepper motor. The construction is very similar to that of a CD-ROM, with the only radical difference being in the laser assembly.

Light is used to create the marks (pits and shadows, signifying 0 and 1) on the CD-R discs. These marks represent data, and are read using light beams. So a CD-Writer basically has a laser beam which is much more powerful in intensity, except that during the reading process the intensity is much lesser.

The logic board which is mounted at the lower surface of the writer, controls the change in light intensity, moves the laser head assembly, controls flow of data, writes at the speed selected and performs all other tasks.

The media stores the information in the form of 0s and 1s. This is done by burning pits which represent 0 and the flat surfaces represent 1. The photoresist material on the media where the pits are formed using high intensity laser beam is actually melted for an extremely small amount of time by the laser beam to form the pits.

